The Vendi–LCR inequality at every order
MathDB 371345

Principia Math In Progress

18 September 2026 (UTC)

Scope and prior work. The current source [1], version 4, already proves the comparison at every nonnegative order, and at negative orders for invertible matrices. The remaining case is a singular matrix at a negative order. We resolve that case using the orthogonal projection onto the range, and give a complete proof for all orders. The positive spectrum convention is the one used in the source's Appendix C.3 code (p. 15). The audit is by the authoring agent, not an independent peer review.

Task 1: full comparison, including singular negative orders

Let $Z = (z_{ij})_{i,j=1}^{n}$ be real symmetric positive semidefinite, with $z_{ii} = 1$ and $0 \le z_{ij} \le 1$. Let $\lambda_1, \dots, \lambda_r > 0$ be its positive eigenvalues, $r = \operatorname{rank} Z$, and put

$$r_i = \sum_j z_{ij}, \qquad s_i = \sum_j z_{ij}^2, \qquad 1 \le s_i \le r_i.$$

For finite $q \neq 1$, define

$$\mathrm{VS}_q(Z) = \left(n^{-q} \sum_{\ell=1}^{r} \lambda_\ell^q \right)^{1/(1-q)}, \qquad D_q(Z) = \left(n^{-q} \sum_{i=1}^{n} r_i^{q-1} \right)^{1/(1-q)}.$$

Thus zero eigenvalues are omitted, including at nonpositive orders. At $q = 1$ use the continuous limits

$$\mathrm{VS}_1 = \exp\left(\log n - \frac{1}{n} \sum_{\ell} \lambda_\ell \log \lambda_\ell \right), \qquad D_1 = \exp\left(\log n - \frac{1}{n} \sum_{i} \log r_i \right).$$

At the two infinite endpoints use the limits in q:

$$\mathrm{VS}_{-\infty} = \frac{n}{\min_\ell \lambda_\ell}, \quad D_{-\infty} = \frac{n}{\min_i r_i}, \qquad \mathrm{VS}_{+\infty} = \frac{n}{\max_\ell \lambda_\ell}, \quad D_{+\infty} = \frac{n}{\max_i r_i}.$$

Theorem 1. *For every such Z and every $q \in [-\infty, +\infty]$, $\mathrm{VS}_q(Z) \ge D_q(Z)$. More precisely, if P is the orthogonal projection onto* $\operatorname{ran} Z$ *and $h_i = P_{ii}$, then for every finite $q \le 0$,*

$$\sum_{\ell=1}^{r} \lambda_\ell^q \;\ge\; \sum_i h_i^{1-q} \;\ge\; \sum_i s_i^{q-1} \;\ge\; \sum_i r_i^{q-1}. \tag{1}$$

Equality $\mathrm{VS}_q = D_q$ at any finite $q \le 0$ holds exactly when Z is, up to simultaneous row and column permutation, a direct sum of all-ones matrices. Such matrices give equality at every order.

The new case: every nonpositive order and arbitrary rank. Since Ze_i lies in $\operatorname{ran} Z$,

$$1 = \langle e_i, Ze_i\rangle = \langle Pe_i, Ze_i\rangle, \qquad 1 \le \|Pe_i\|^2\|Ze_i\|^2 = h_i s_i.$$

In particular $h_i > 0$ and $h_i \ge 1/s_i \ge 1/r_i$. Choose an orthonormal eigenbasis $v_1, \dots, v_r$ for the range and write $w_{i\ell} = |(v_\ell)_i|^2$. Then

$$\sum_\ell w_{i\ell} = h_i, \qquad \sum_\ell \lambda_\ell w_{i\ell} = z_{ii} = 1.$$

For $q < 0$, the function $t \mapsto t^q$ is convex on $(0, \infty)$. Jensen's inequality with weights $w_{i\ell}/h_i$ gives

$$\sum_\ell w_{i\ell}\lambda_\ell^q \ge h_i \left(\frac{1}{h_i}\right)^q = h_i^{1-q}.$$

Summing over i proves the first inequality in (1), since $\sum_i w_{i\ell} = 1$. For $q = 0$ that first inequality is equality: $r = \sum_i h_i$. The remaining inequalities follow from $h_i \ge 1/s_i$ and $s_i \le r_i$, with $1 - q > 0$ and $q - 1 < 0$. Multiplication by n^{-q} and raising to the positive power $1/(1-q)$ proves the score comparison for all finite $q \le 0$.

For equality, the last inequality in (1) must be equality term by term. Hence $s_i = r_i$ for every i, and $\sum_j z_{ij}(1 - z_{ij}) = 0$ forces every entry to be either zero or one. Represent Z as a Gram matrix of unit vectors. An inner product of one means two vectors are equal; an inner product of zero means they are orthogonal. The equal-vector classes therefore give the claimed all-ones blocks. Conversely, if the block sizes are $b_1, \dots, b_r$, the positive eigenvalues are $b_1, \dots, b_r$ and

$$\sum_\ell \lambda_\ell^q = \sum_\ell b_\ell^q = \sum_i r_i^{q-1}.$$

The equality conclusion follows, including the limits at 1 and infinity. □

Completion at positive orders (already established in [1]). For completeness, we give another proof of this known portion. Set $R = \operatorname{diag}(r_1, \dots, r_n)$. For every vector u,

$$u^*(R - Z)u = \frac{1}{2}\sum_{i,j} z_{ij}|u_i - u_j|^2 \ge 0, \qquad 0 \preceq Z \preceq R.$$

We use the familiar matrix power-order facts

$$A \preceq B \Longrightarrow A^\alpha \preceq B^\alpha \quad (0 < \alpha \le 1), \qquad A^{-p} \succeq B^{-p} \quad (0 < p \le 1, A \succ 0).$$

They follow from the reversal of order under inversion and the scalar integrals, applied by spectral calculus,

$$x^\alpha = \frac{\sin \pi\alpha}{\pi}\int_0^\infty t^{\alpha-1}\frac{x}{x+t}\,dt, \qquad x^{-p} = \frac{\sin \pi p}{\pi}\int_0^\infty \frac{t^{-p}}{x+t}\,dt$$

for exponents strictly between zero and one; the endpoint cases are direct. Inversion reverses order because conjugating $A \preceq B$ by $B^{-1/2}$ gives a positive definite matrix bounded above by the identity, whose inverse is bounded below by the identity.

For $1 < q \le 2$, take $\alpha = q - 1$. Multiplying the power-order inequality by Z inside a trace gives

$$\operatorname{tr} Z^q \le \operatorname{tr}(ZR^{q-1}) = \sum_i r_i^{q-1}.$$

For $q \geq 2$, use a unit eigenvector v_ℓ of Z and scalar Jensen:

$$\langle v_\ell, R^{q-1} v_\ell \rangle \geq \langle v_\ell, R v_\ell \rangle^{q-1} \geq \lambda_\ell^{q-1}.$$

Multiplying by λ_ℓ and summing gives the same trace inequality. Because $1/(1-q) < 0$, it proves $\mathrm{VS}_q \geq D_q$ for $q > 1$.

For $0 < q < 1$, put $p = 1 - q$. Apply the inverse-power order to $Z + \varepsilon I \preceq R + \varepsilon I$, multiply by Z inside the trace, and let $\varepsilon \downarrow 0$:

$$\sum_{\ell=1}^{r} \lambda_\ell^q = \lim_{\varepsilon \downarrow 0} \mathrm{tr}\big(Z(Z+\varepsilon I)^{-p}\big) \geq \mathrm{tr}(ZR^{-p}) = \sum_i r_i^{q-1}.$$

The zero-eigenvalue contributions are identically zero before the limit; this is a finite-dimensional spectral limit. The positive exponent $1/(1-q)$ now gives $\mathrm{VS}_q \geq D_q$. Continuity gives $q = 1$. Both spectra and the list of row sums are finite and strictly positive after removing zero eigenvalues. Taking $q \to -\infty$ and $q \to +\infty$ therefore gives the two endpoint formulas and the corresponding inequalities. This completes the proof of Theorem 1.

Why the singular case needs its own argument. The full-rank estimate $\mathrm{VS}_q \geq n \geq D_q$ at negative orders does not apply when $\operatorname{rank} Z < n$. Nor can zero eigenvalues simply be replaced by small positive numbers: negative powers then diverge, whereas the intended score sums only over the positive spectrum. Formula (1) works directly on that spectrum, with no such limiting substitution. If instead one adopts the convention that zero eigenvalues contribute $+\infty$ at negative order, the comparison is immediate there; the theorem above proves the stronger, finite, source-intended version.

Tasks 2–5: proposed follow-up problems

The following are directions, not results proved in this note.

Task 2. Extend the full comparison to a nonuniform probability vector p, comparing the spectrum of $\mathrm{diag}(\sqrt{p})Z\,\mathrm{diag}(\sqrt{p})$ with $\big(\sum_i p_i (Zp)_i^{q-1}\big)^{1/(1-q)}$, with appropriate support conventions.

Task 3. Classify all equality cases at each finite positive order, including the Shannon order $q = 1$.

Task 4. At a fixed $q < 0$, obtain optimal quantitative stability bounds relating the trace gap in (1) to $\sum_{i,j} z_{ij}(1 - z_{ij})$ and distance from all-ones block matrices.

Task 5. For prescribed n, rank, $q < 0$, and a lower bound $\varepsilon > 0$ on the positive eigenvalues, determine the maximum possible ratio $\mathrm{VS}_q(Z)/D_q(Z)$ and characterize extremizers.

References

[1] P. Nguyen, J. Couch, R. Bansal, A. Morgan, C. Tam, M. Li, Rima Arnaout, and Ramy Arnaout, *Which Similarity-Sensitive Entropy (Sentropy)?*, arXiv:2511.03849v4, submitted 30 March 2026, PDF dated 1 April 2026. Theorem 1 and Section III, pp. 3–6; Appendix C.3, p. 15. `https://arxiv.org/abs/2511.03849v4`.